# GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of seed mass per plot (esy4)

MB

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## Contents



> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

## Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```r
library(DHARMa)
library(emmeans)
```

**Packages**

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
  mutate(
    esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
    eco.id = factor(eco.id)
    ) %>%
  rename(y = cwm.abu.seedmass)
```

**Load data**

# Statistics

## Data exploration

### Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##       upper        mean       lower
## 0.001574195 0.001489662 0.001405129
```

```r
median(sites$y)
```

```
## [1] 0.00122
```

```r
sd(sites$y)
```

```
## [1] 0.001072695
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%     95%
## 0.00038 0.00347
```

```r
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id     n
```

```
##   <fct>  <int>
## 1 654      203
## 2 664      203
## 3 686      215
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive    102
## 2 restored    401
## 3 negative    118
```

```r
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R       330
## 2 R22     210
## 3 R1A      81
```

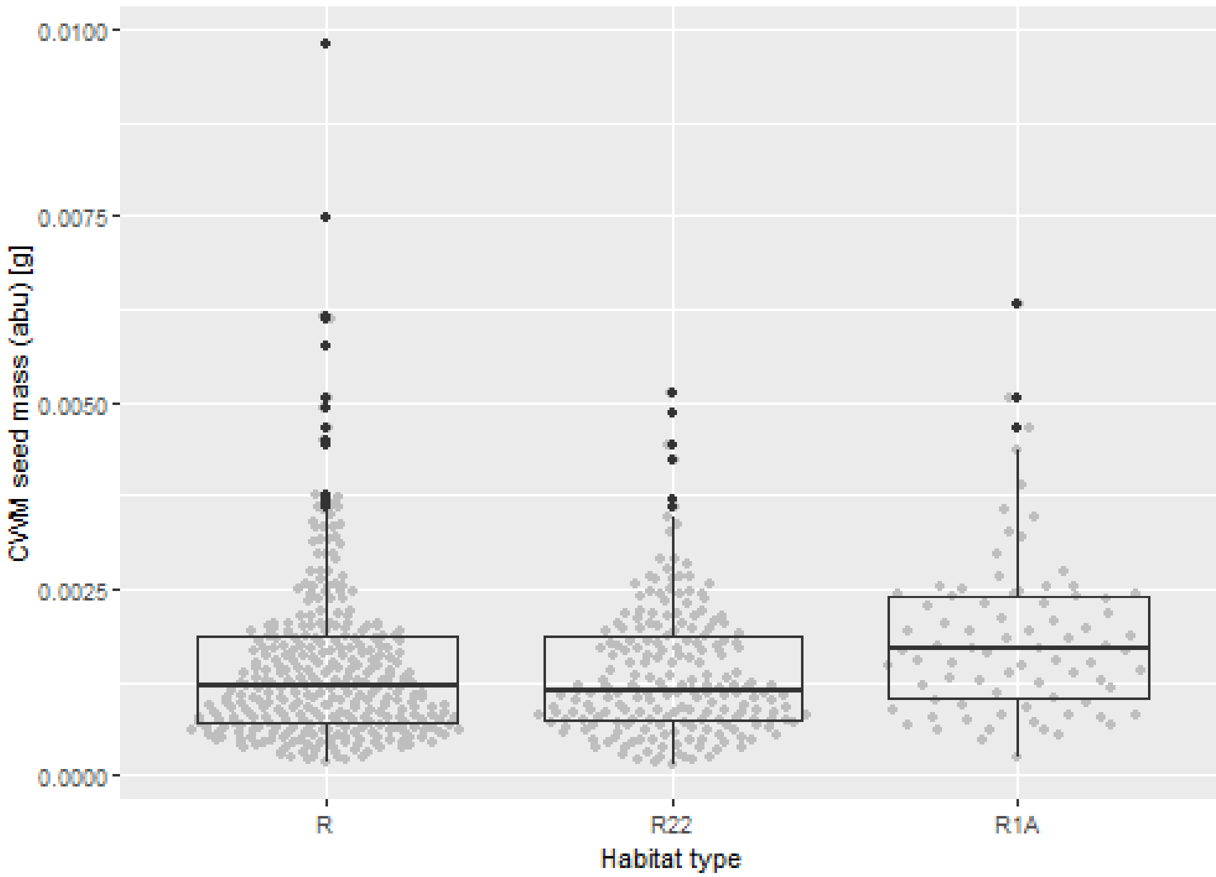
```r
sites %>% count(esy4, eco.id)
```

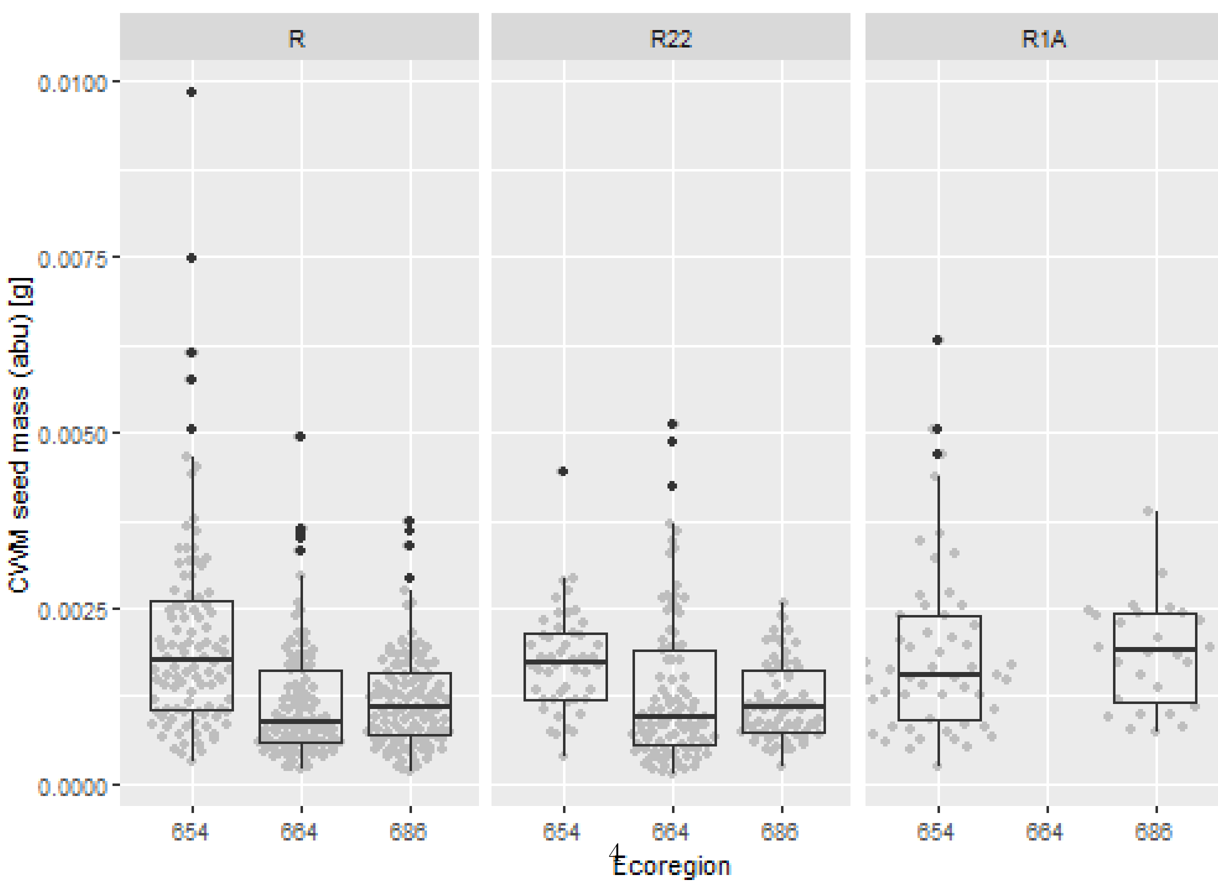
```
## # A tibble: 8 x 3
##   esy4  eco.id     n
##   <fct> <fct>  <int>
## 1 R     654      102
## 2 R     664      112
## 3 R     686      116
## 4 R22   654       48
## 5 R22   664       91
## 6 R22   686       71
## 7 R1A   654       53
## 8 R1A   686       28
```

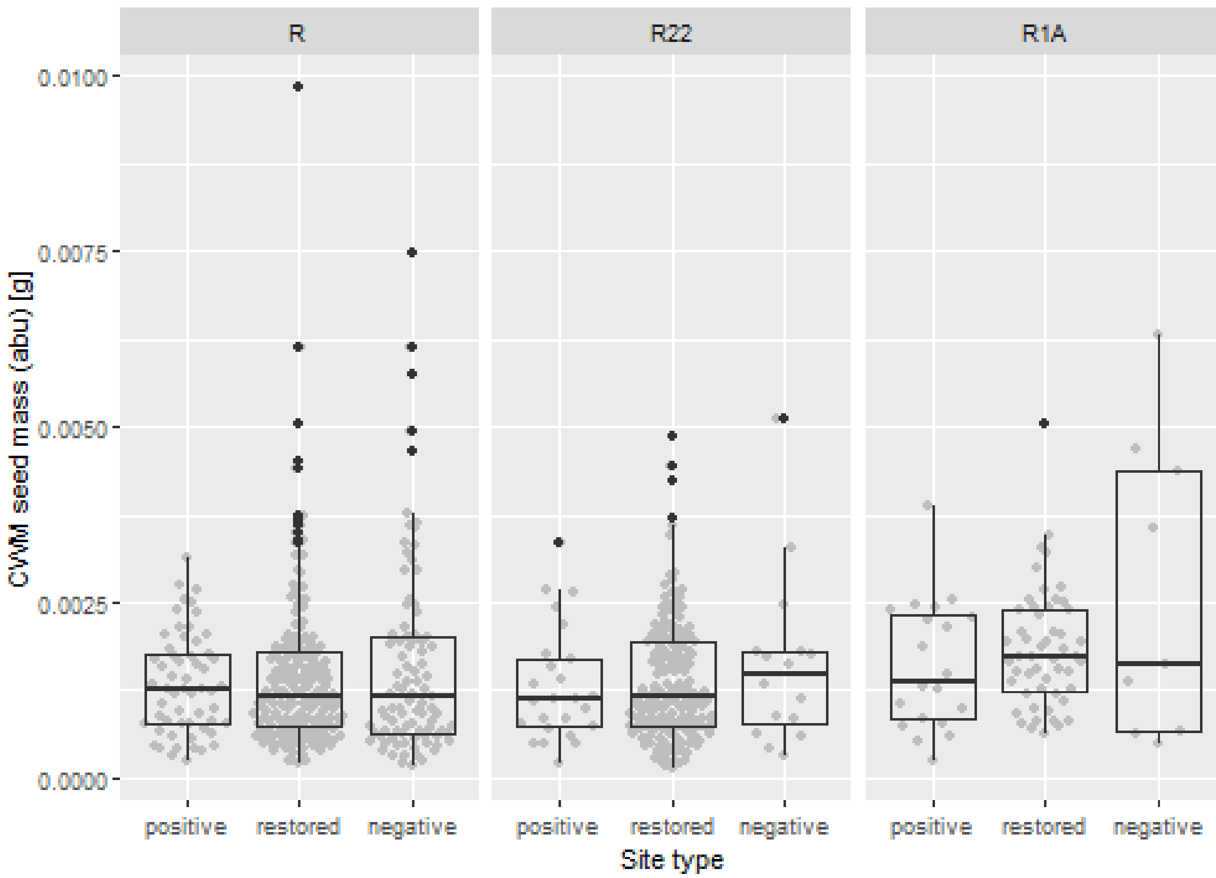
```r
sites %>% count(esy4, site.type)
```

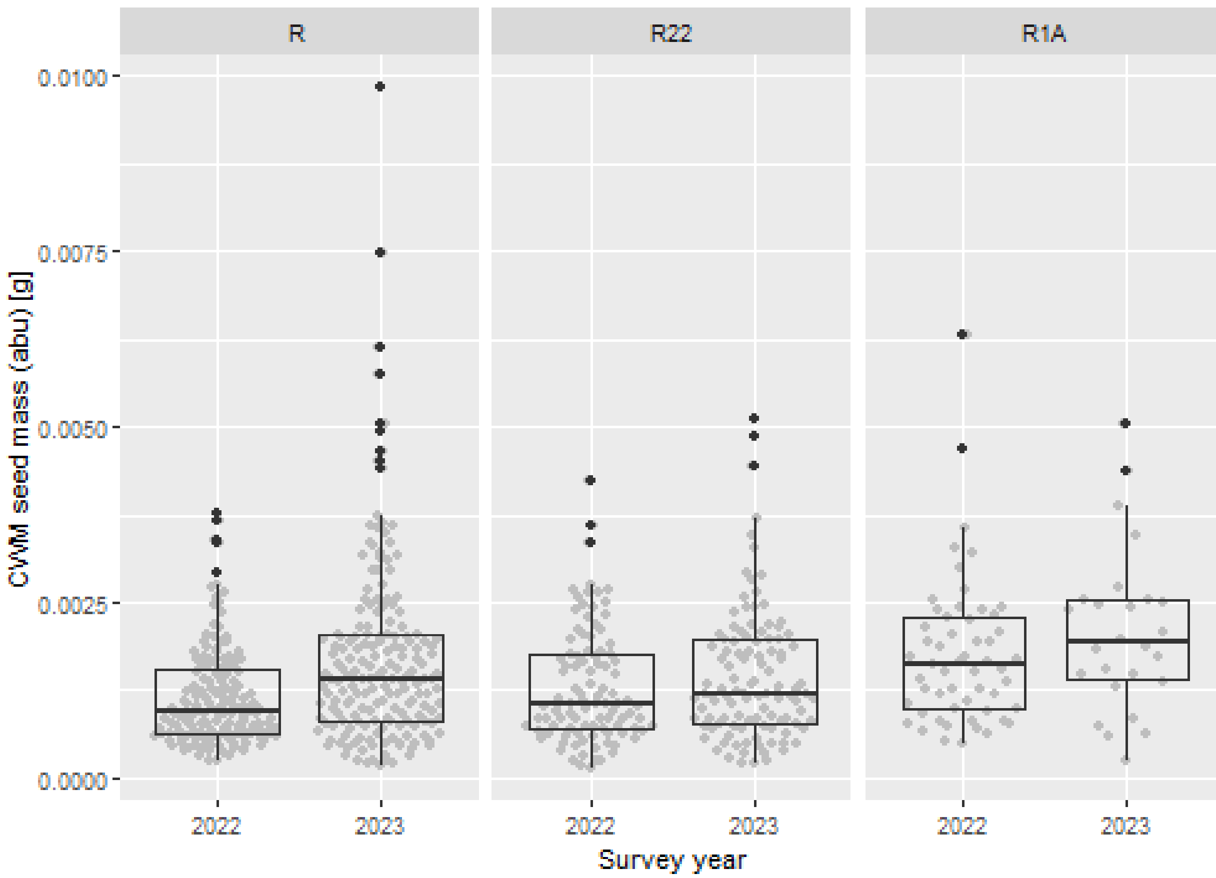
```
## # A tibble: 9 x 3
##   esy4  site.type     n
##   <fct> <ord>     <int>
## 1 R     positive     57
## 2 R     restored    180
## 3 R     negative     93
## 4 R22   positive     25
## 5 R22   restored    169
## 6 R22   negative     16
## 7 R1A   positive     20
## 8 R1A   restored     52
## 9 R1A   negative      9
```

**Graphs of raw data (Step 2, 6, 7)**



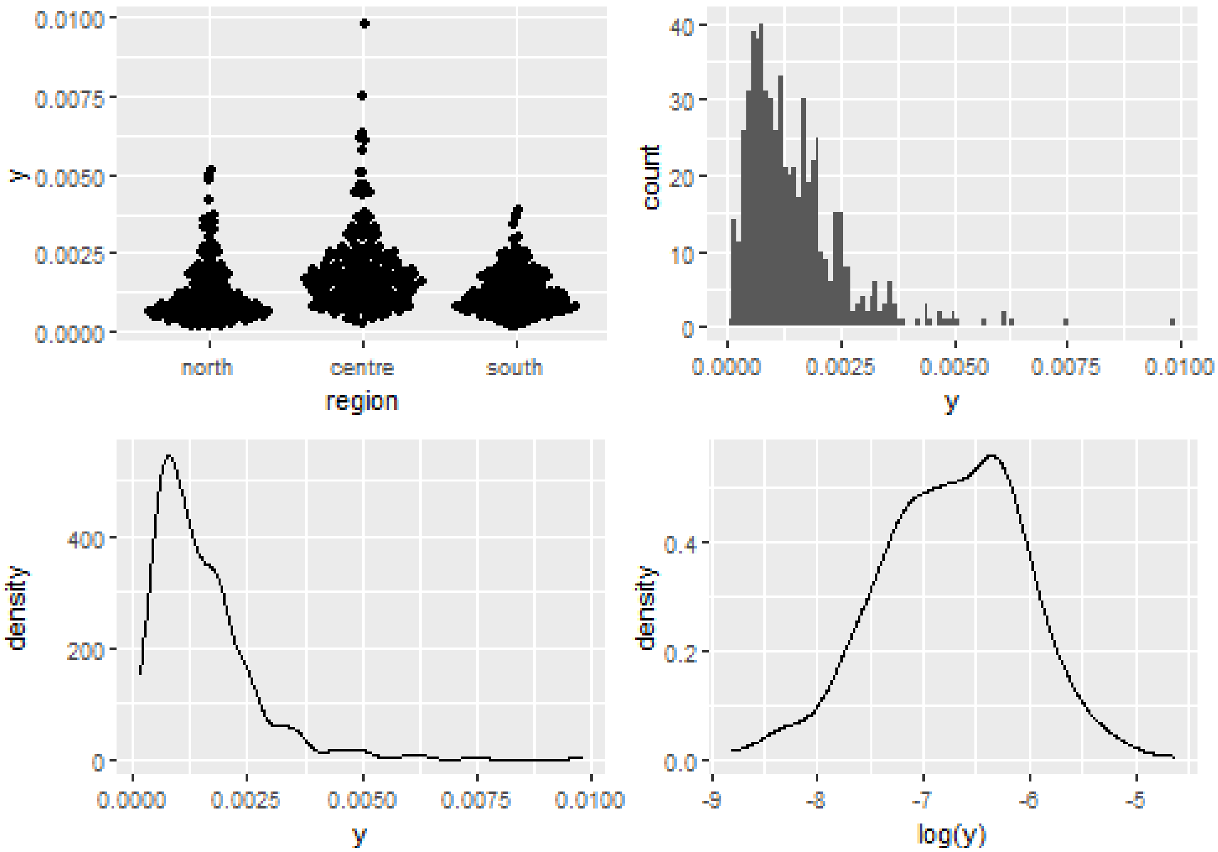




R
R22
R1A
0.0100
0.0075
0.0050
0.0025
0.0000
CWM seed mass (abu) [g]
positive
restored
negative
positive
restored
negative
positive
restored
negative
Site type



R
R22
R1A
0.0100
0.0075
0.0050
0.0025
0.0000
CWM seed mass (abu) [g]
2022
2023
2022
2023
2022
2023
Survey year

**Outliers, zero-inflation, transformations? (Step 1, 3, 4)**




**Check collinearity part 1 (Step 5)**

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

## Models

**NOTE:** Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_seedmass_esy4_1.Rdata"))
load(file = here("outputs", "models", "model_seedmass_esy4_3.Rdata"))
m_1 <- m1
m_2 <- m3
```

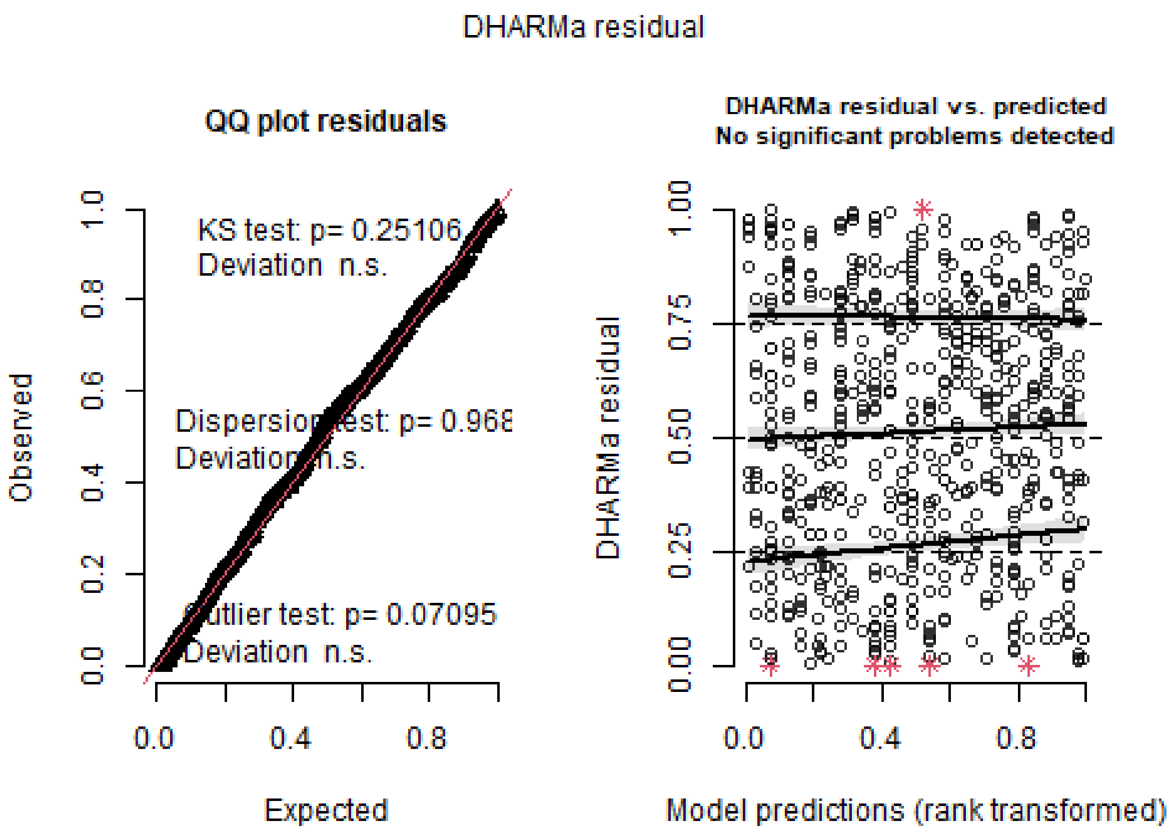
```
m_1@call
## lmer(formula = log(y) ~ esy4 * (site.type + eco.id) + obs.year +
##     (1 | id.site), data = sites, REML = FALSE)
```

```r
m_2@call
## lmer(formula = log(y) ~ esy4 * (site.type + eco.id) + obs.year +
##     hydrology + (1 | id.site), data = sites, REML = FALSE)
```

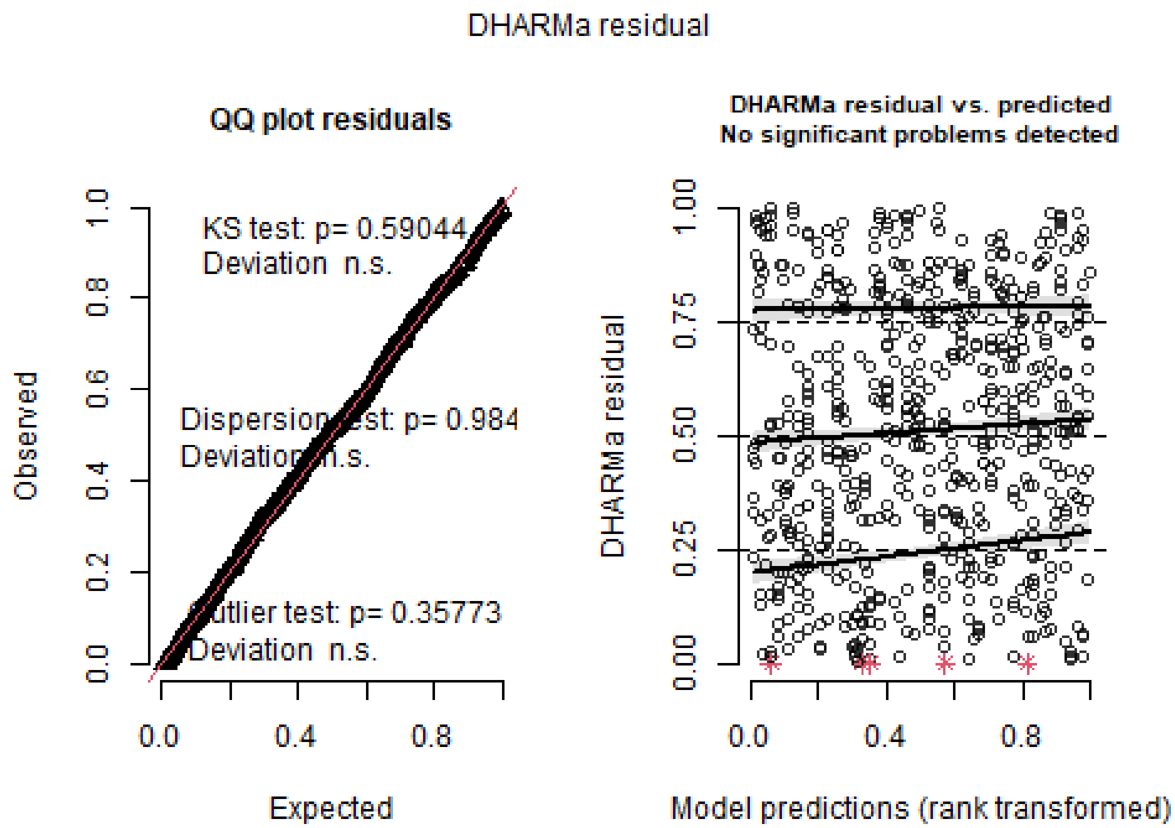
## Model check

### DHARMa

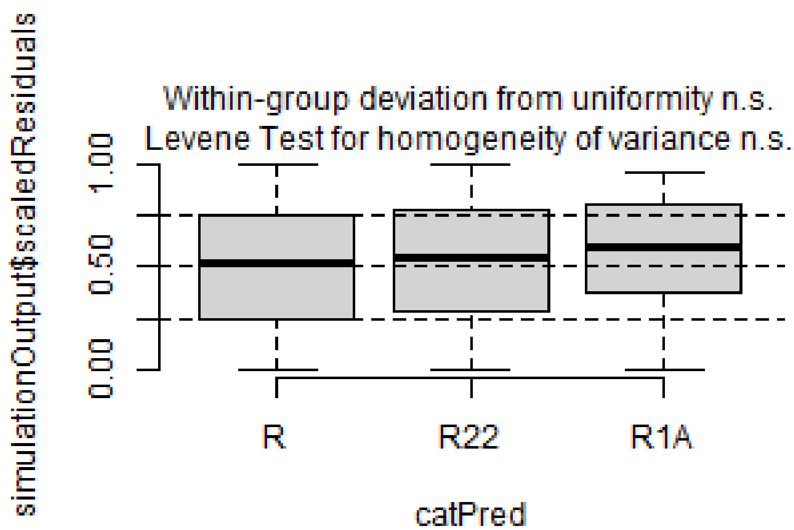
```r
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



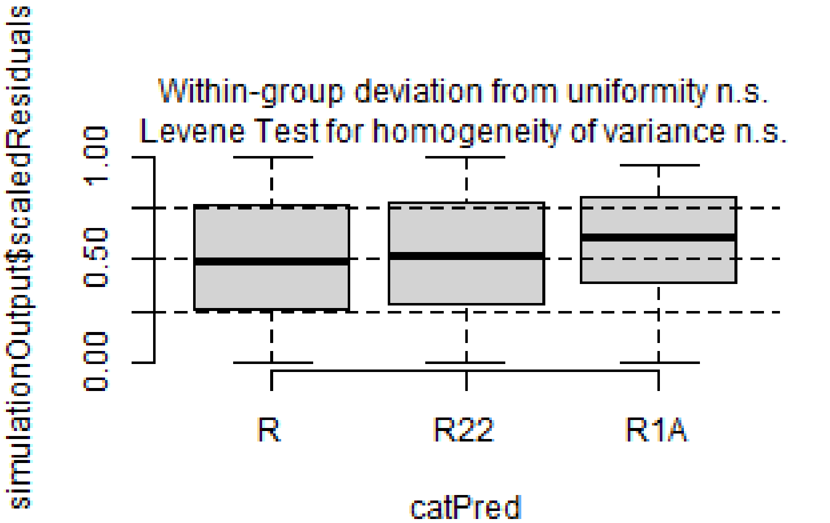

```r
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



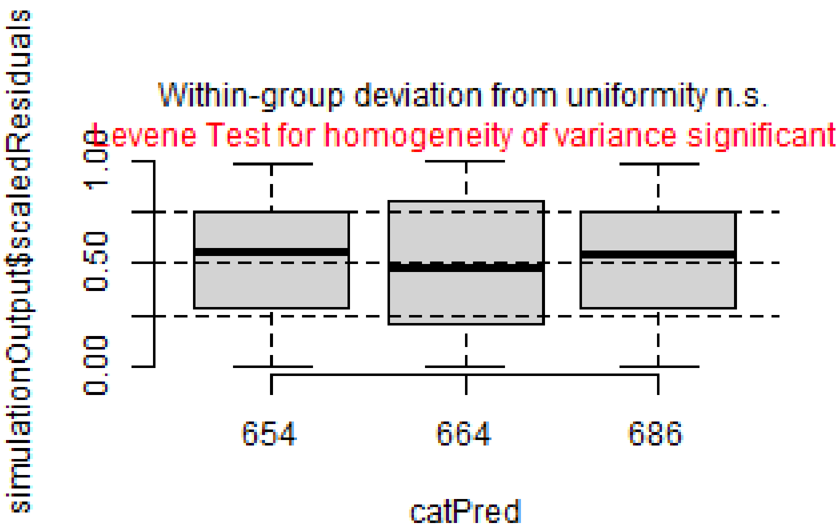

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



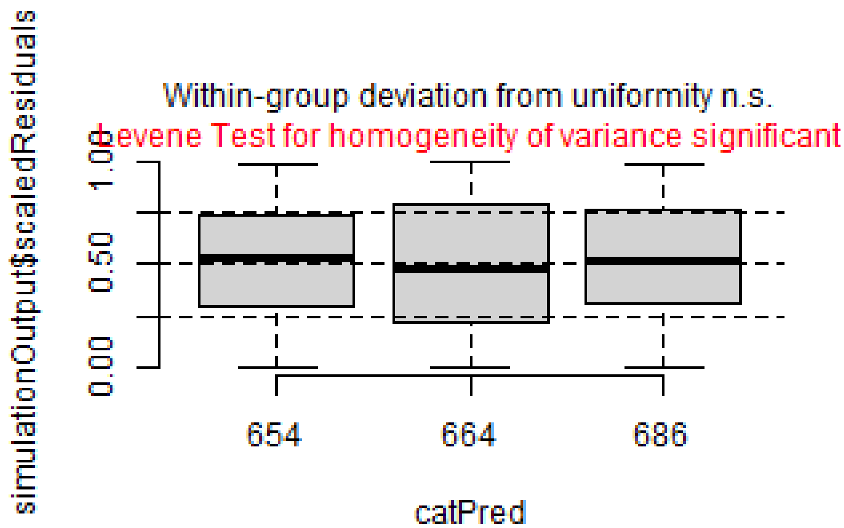

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



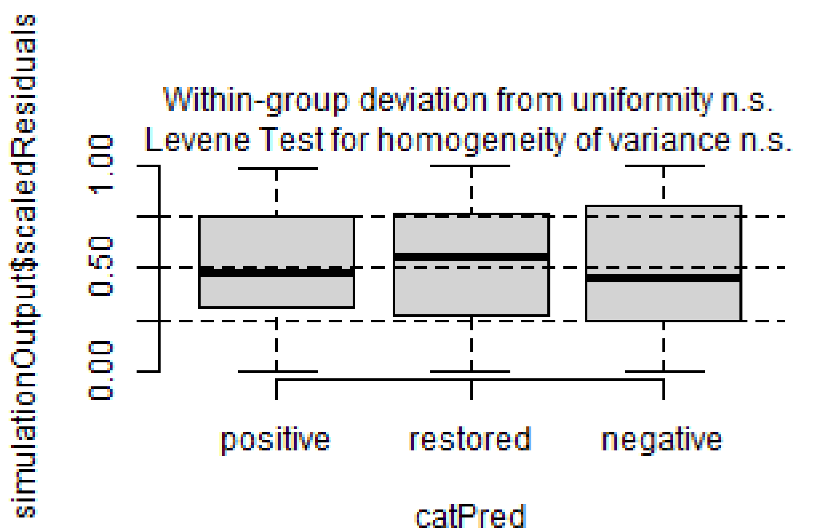

```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



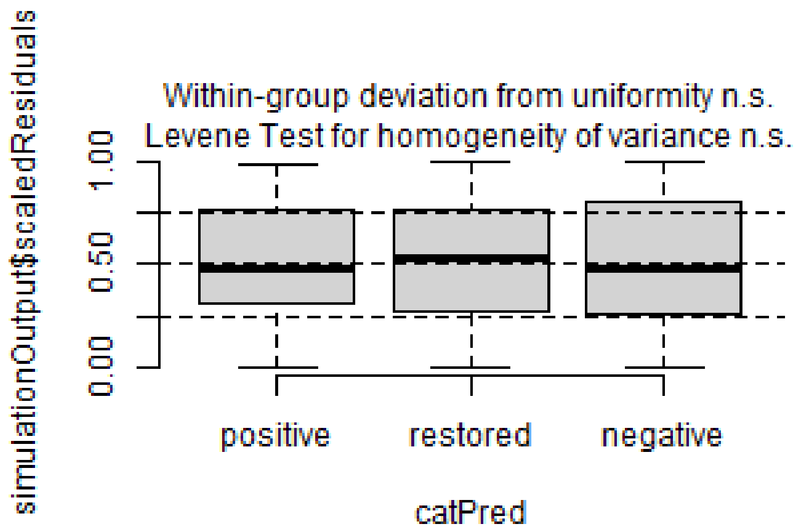

```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



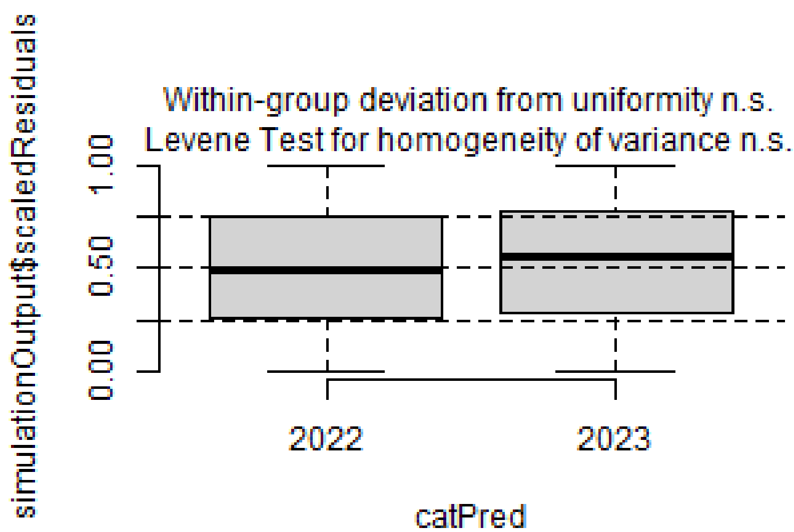

```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



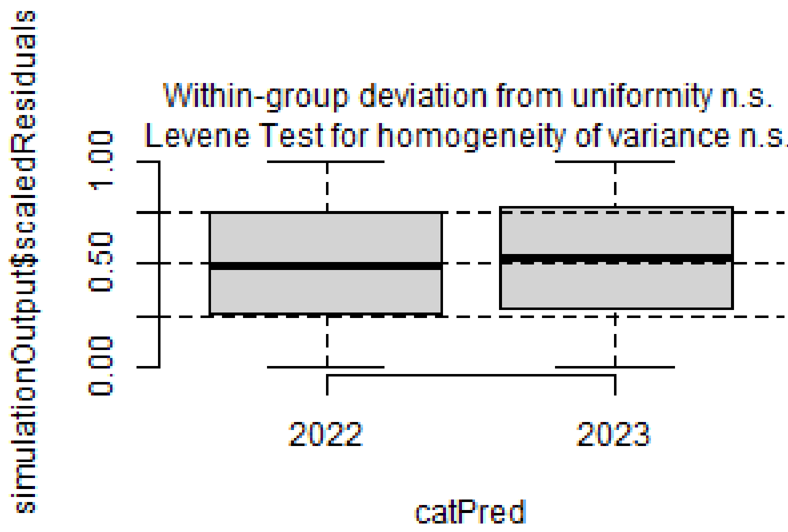

```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



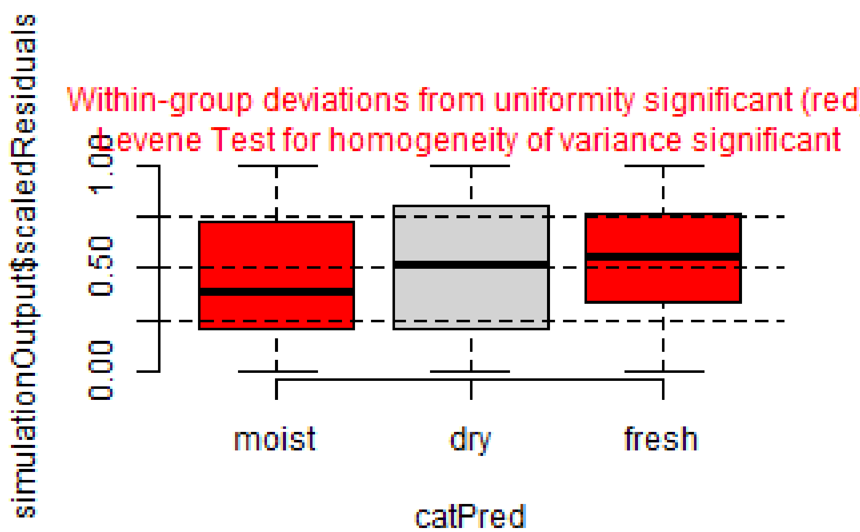

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



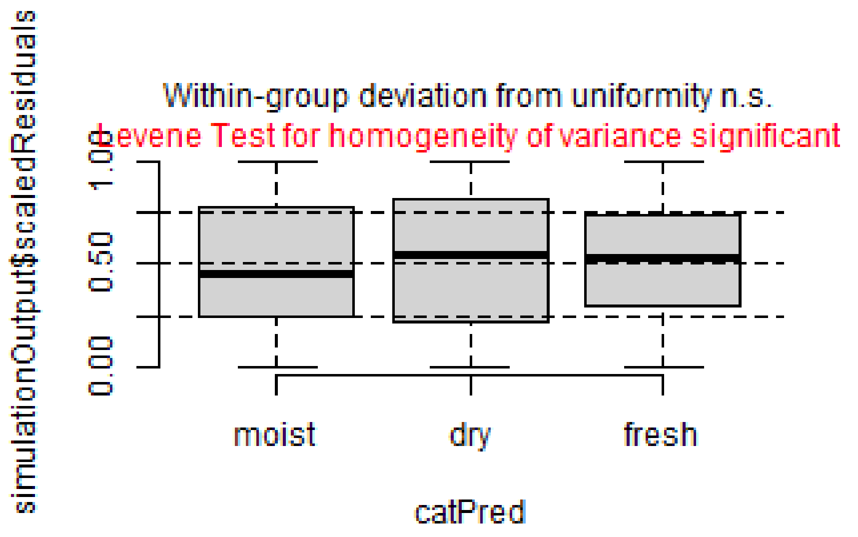

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



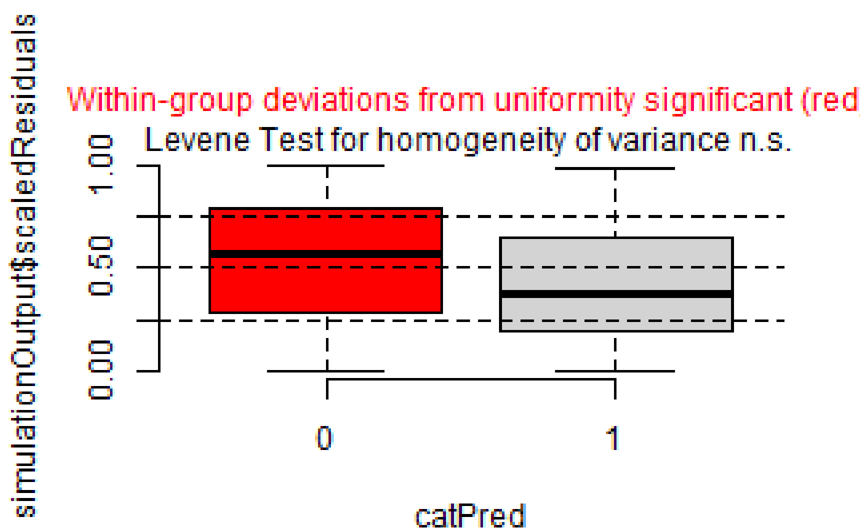

```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



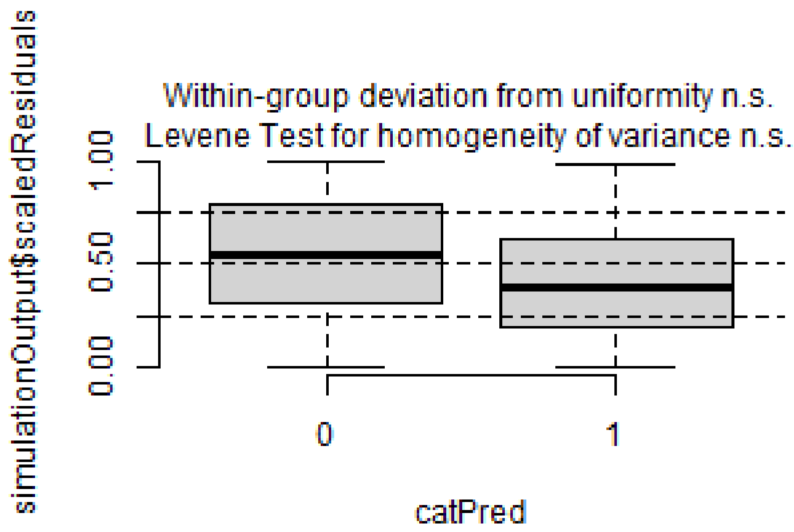

```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



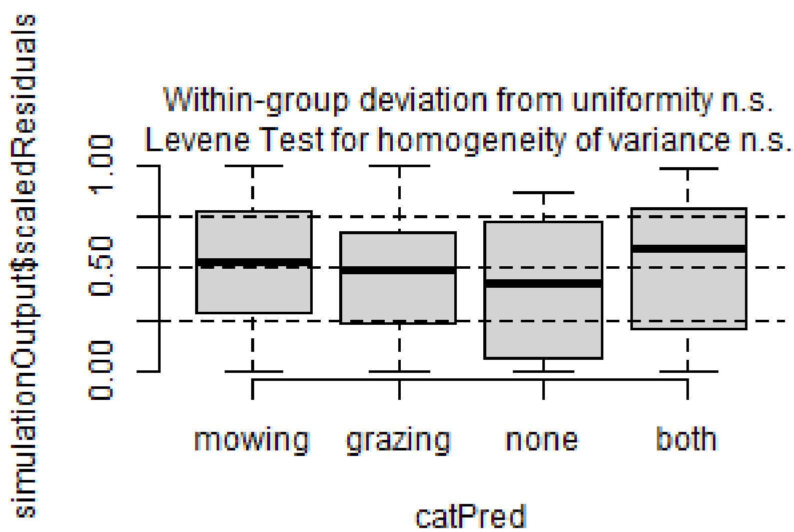

```
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



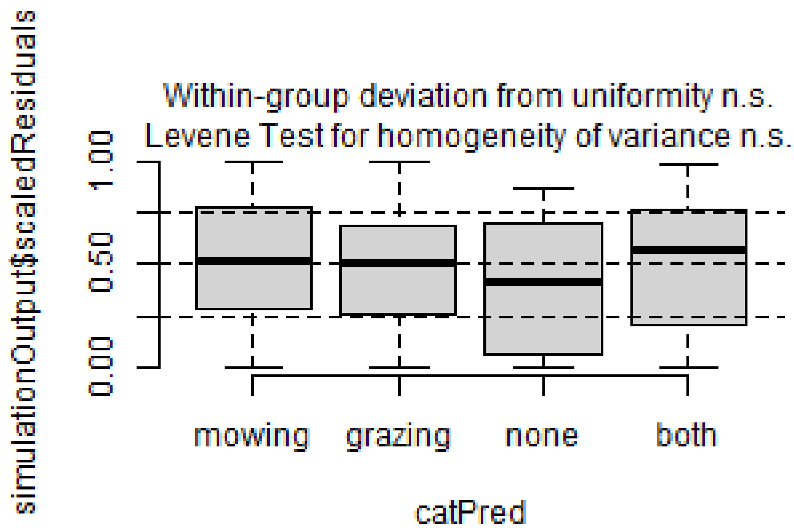

```
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```




```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```




### Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##                     GVIF Df GVIF^(1/(2*Df))
## esy4           11.330679  2        1.834695
## site.type       1.418765  2        1.091384
## eco.id          1.594355  2        1.123689
## obs.year        1.022266  1        1.011072
## esy4:site.type  4.999271  4        1.222822
## esy4:eco.id     9.139388  3        1.445949
```

```
car::vif(m_2)
```

```
##                     GVIF Df GVIF^(1/(2*Df))
## esy4           12.586319  2        1.883539
## site.type       1.504139  2        1.107445
## eco.id          1.630581  2        1.130019
## obs.year        1.029409  1        1.014598
## hydrology       1.300347  2        1.067861
## esy4:site.type  5.063599  4        1.224778
## esy4:eco.id     9.354352  3        1.451562
```

## Model comparison

### R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.1409498 0.6805522
MuMIn::r.squaredGLMM(m_2)
##           R2m       R2c
## [1,] 0.149761 0.6790846
```

**AICc**

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_1 17 955.5114
## m_2 19 956.6572
```

## Predicted values

### Summary table

```
car::Anova(m_2, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: log(y)
##                     Chisq Df Pr(>Chisq)
## (Intercept)    2877.1107  1  < 2.2e-16 ***
## esy4              0.5213  2    0.77055
## site.type         0.7991  2    0.67062
## eco.id           23.2769  2   8.82e-06 ***
## obs.year          5.2836  1    0.02153 *
## hydrology         3.1367  2    0.20839
## esy4:site.type    3.4804  4    0.48087
## esy4:eco.id       3.8800  3    0.27471
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_2)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: log(y) ~ esy4 * (site.type + eco.id) + obs.year + hydrology +
##     (1 | id.site)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     955.4    1039.6    -458.7     917.4       602
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9928 -0.4799  0.0140  0.4601  2.8670
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  id.site  (Intercept) 0.2472   0.4972
##  Residual             0.1499   0.3871
## Number of obs: 621, groups:  id.site, 177
##
## Fixed effects:
##                      Estimate Std. Error t value
## (Intercept)          -6.55120    0.12214 -53.639
## esy4R22              -0.04730    0.10199  -0.464
## esy4R1A               0.07028    0.14104   0.498
```

```
## site.type.L         -0.09094   0.10684  -0.851
## site.type.Q         -0.01542   0.07743  -0.199
## eco.id664           -0.52268   0.11482  -4.552
## eco.id686           -0.41497   0.11180  -3.712
## obs.year2023         0.19000   0.08266   2.299
## hydrologyfresh       0.10291   0.10752   0.957
## hydrologymoist      -0.07868   0.12397  -0.635
## esy4R22:site.type.L  0.15406   0.12845   1.199
## esy4R1A:site.type.L  0.36780   0.24447   1.504
## esy4R22:site.type.Q  0.02517   0.08767   0.287
## esy4R1A:site.type.Q  0.12844   0.16818   0.764
## esy4R22:eco.id664    0.03455   0.12635   0.273
## esy4R22:eco.id686   -0.03764   0.11714  -0.321
## esy4R1A:eco.id686    0.32621   0.18321   1.781
```

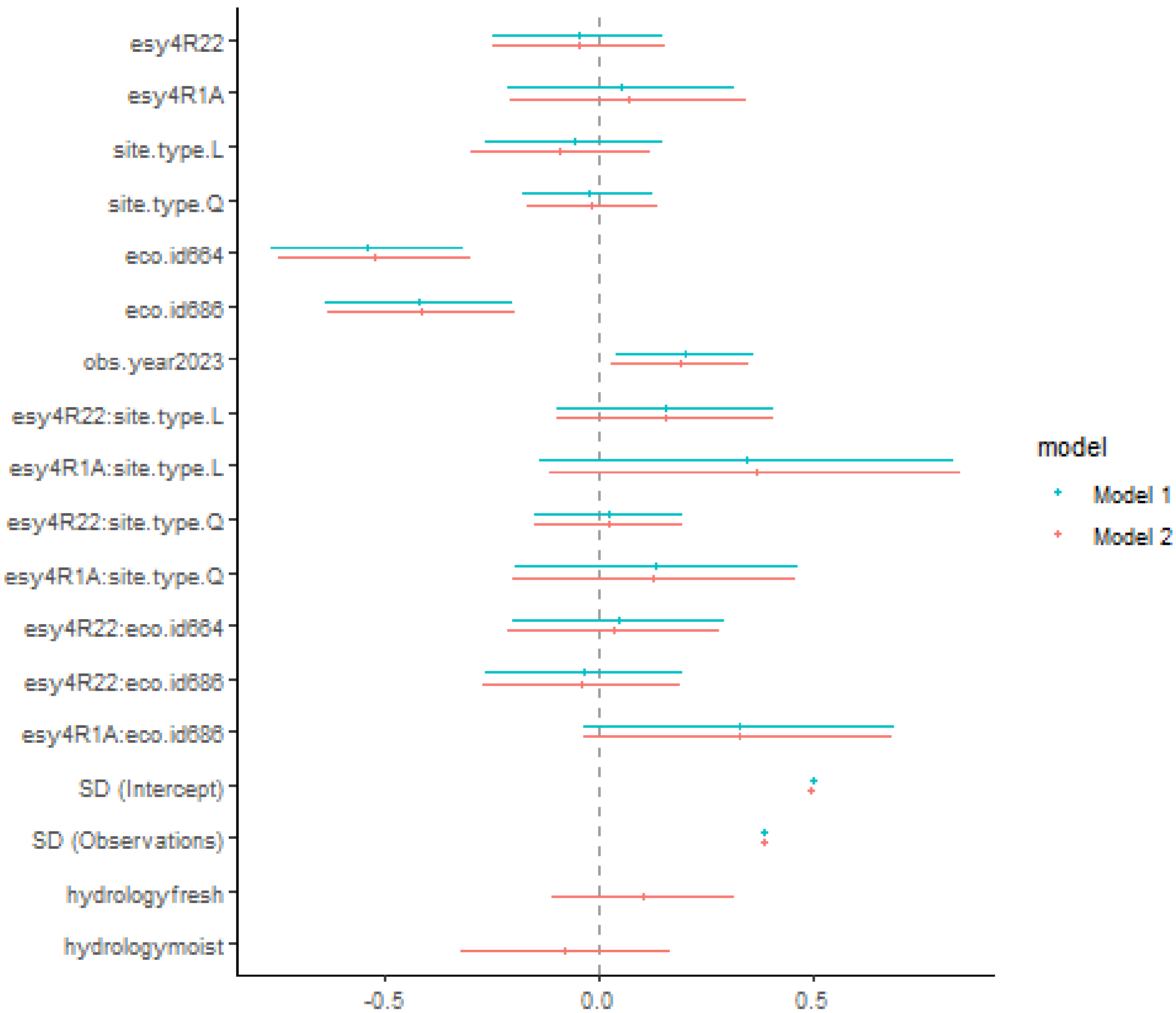
```
##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

**Forest plot**

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```




### Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```r
(emm <- emmeans(m_2, "esy4", type = "response"))
```

### ESy EUNIS Habitat type

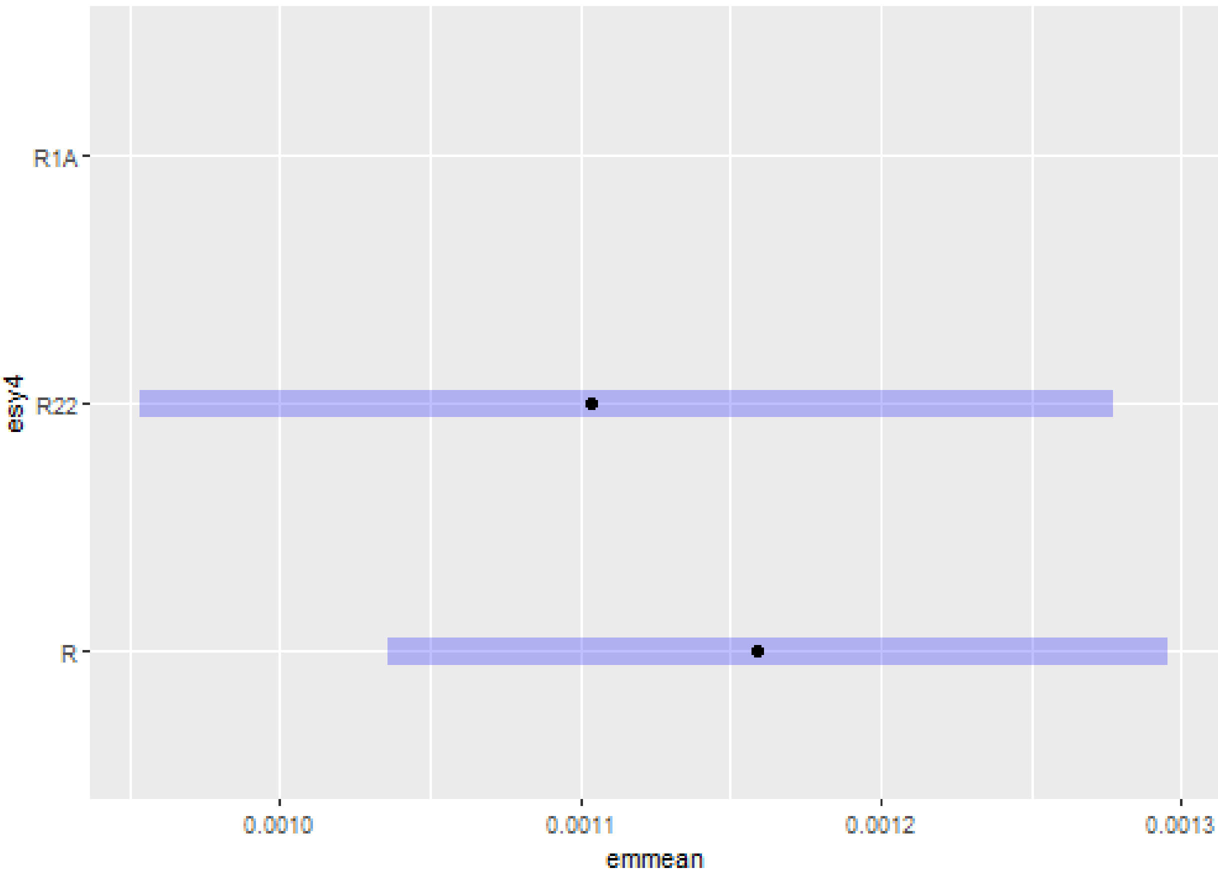
```
##  esy4 response       SE  df lower.CL upper.CL
##  R     0.00116 6.58e-05 156 0.001036  0.00130
##  R22   0.00110 8.19e-05 289 0.000954  0.00128
##  R1A    nonEst       NA  NA       NA       NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
```

```r
plot(emm, comparison = FALSE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
(emm <- emmeans(
  m_2,
  revpairwise ~ eco.id + esy4,
  type = "response"
  ))
```

**Habiat type x Region**

```
## $emmeans
##  eco.id esy4 response       SE  df lower.CL upper.CL
##  654    R    0.001583 1.41e-04 200 0.001329  0.00189
##  664    R    0.000939 8.30e-05 215 0.000789  0.00112
##  686    R    0.001046 8.99e-05 183 0.000883  0.00124
##  654    R22  0.001510 1.75e-04 374 0.001202  0.00190
##  664    R22  0.000927 9.27e-05 298 0.000761  0.00113
##  686    R22  0.000961 9.86e-05 296 0.000785  0.00118
##  654    R1A  0.001699 2.35e-04 220 0.001294  0.00223
```

```
##  664    R1A    nonEst       NA  NA       NA       NA
##  686    R1A  0.001554 2.77e-04 452 0.001095  0.00221
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
##  contrast                       ratio     SE  df null t.ratio p.value
##  eco.id664 R / eco.id654 R      0.593 0.0699 233    1  -4.431  0.0004
##  eco.id686 R / eco.id654 R      0.660 0.0759 247    1  -3.612  0.0087
##  eco.id686 R / eco.id664 R      1.114 0.1280 248    1   0.938  0.9820
##  eco.id654 R22 / eco.id654 R    0.954 0.0985 565    1  -0.458  0.9998
##  eco.id654 R22 / eco.id664 R    1.609 0.2250 385    1   3.394  0.0172
##  eco.id654 R22 / eco.id686 R    1.444 0.1980 402    1   2.681  0.1315
##  eco.id664 R22 / eco.id654 R    0.585 0.0749 332    1  -4.183  0.0010
##  eco.id664 R22 / eco.id664 R    0.987 0.0954 587    1  -0.132  1.0000
##  eco.id664 R22 / eco.id686 R    0.887 0.1100 326    1  -0.970  0.9784
##  eco.id664 R22 / eco.id654 R22  0.614 0.0835 381    1  -3.589  0.0089
##  eco.id686 R22 / eco.id654 R    0.607 0.0783 352    1  -3.874  0.0032
##  eco.id686 R22 / eco.id664 R    1.023 0.1310 334    1   0.177  1.0000
##  eco.id686 R22 / eco.id686 R    0.919 0.0786 537    1  -0.992  0.9755
##  eco.id686 R22 / eco.id654 R22  0.636 0.0861 393    1  -3.343  0.0203
##  eco.id686 R22 / eco.id664 R22  1.036 0.1300 324    1   0.284  1.0000
##  eco.id654 R1A / eco.id654 R    1.073 0.1540 309    1   0.488  0.9997
##  eco.id654 R1A / eco.id664 R    1.809 0.2980 189    1   3.598  0.0096
##  eco.id654 R1A / eco.id686 R    1.625 0.2650 199    1   2.977  0.0635
##  eco.id654 R1A / eco.id654 R22  1.125 0.1890 325    1   0.701  0.9969
##  eco.id654 R1A / eco.id664 R22  1.833 0.3120 237    1   3.558  0.0105
##  eco.id654 R1A / eco.id686 R22  1.769 0.3050 237    1   3.310  0.0236
##  eco.id664 R1A / eco.id654 R   nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id664 R   nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id686 R   nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id654 R22 nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id664 R22 nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id686 R22 nonEst     NA  NA    1      NA      NA
##  eco.id664 R1A / eco.id654 R1A nonEst     NA  NA    1      NA      NA
##  eco.id686 R1A / eco.id654 R    0.982 0.1940 418    1  -0.093  1.0000
##  eco.id686 R1A / eco.id664 R    1.656 0.3290 378    1   2.537  0.1832
##  eco.id686 R1A / eco.id686 R    1.487 0.2660 490    1   2.213  0.3457
##  eco.id686 R1A / eco.id654 R22  1.029 0.2190 436    1   0.136  1.0000
##  eco.id686 R1A / eco.id664 R22  1.677 0.3410 420    1   2.544  0.1801
##  eco.id686 R1A / eco.id686 R22  1.618 0.3060 515    1   2.546  0.1789
##  eco.id686 R1A / eco.id654 R1A  0.915 0.1700 431    1  -0.477  0.9998
##  eco.id686 R1A / eco.id664 R1A nonEst     NA  NA    1      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 8 estimates
## Tests are performed on the log scale
```

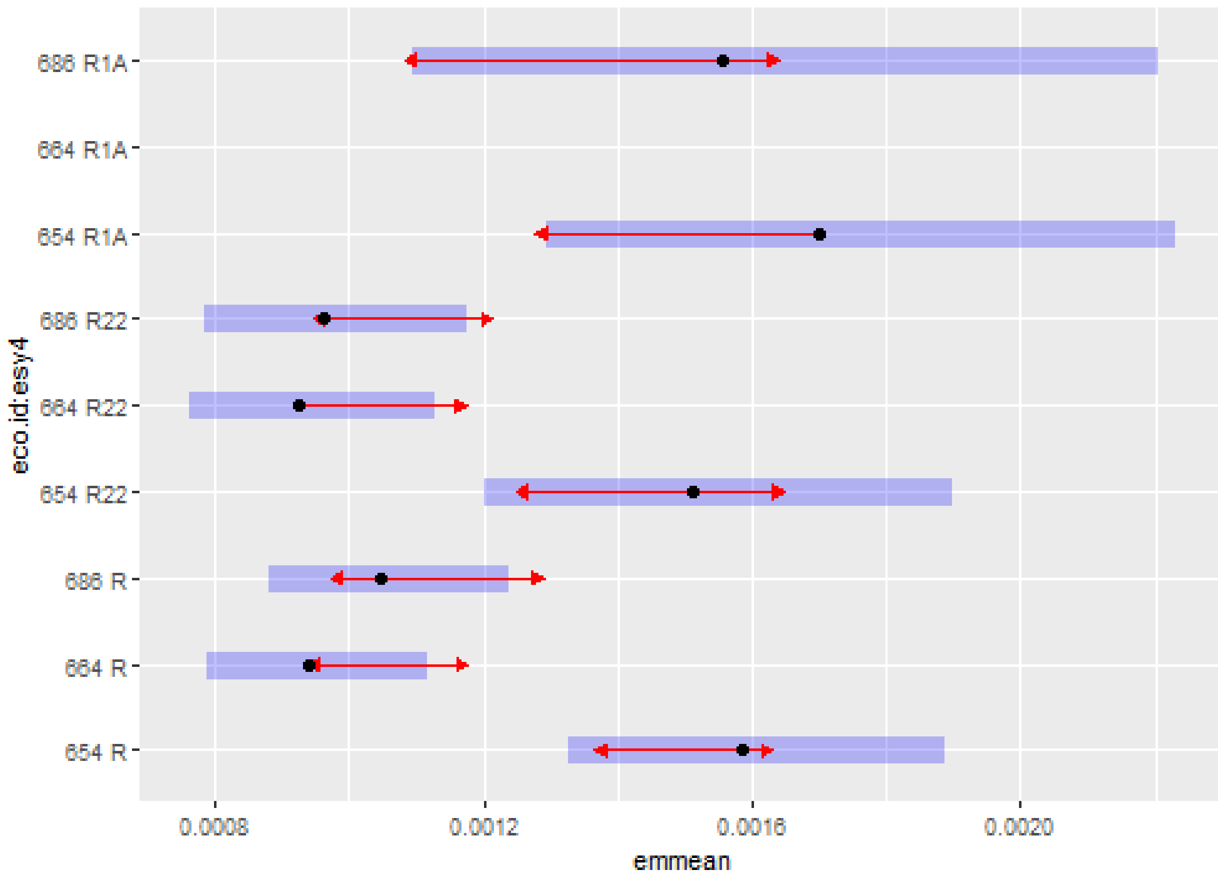
```r
plot(emm, comparison = TRUE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# (emm <- emmeans(
#   m_2,
#   revpairwise ~ site.type | esy4,
#   type = "response"
#   ))
# plot(emm, comparison = TRUE)
```

**Habiat type x Site type**

## Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
```

```
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     patchwork_1.3.0  ggbeeswarm_0.7.2
##  [5] lubridate_1.9.4  forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4
##  [9] purrr_1.0.4      readr_2.1.5      tidyr_1.3.1      tibble_3.2.1
## [13] ggplot2_3.5.2    tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.4           gridExtra_2.3          sandwich_3.1-1
##  [4] rlang_1.1.6            magrittr_2.0.3         multcomp_1.4-28
##  [7] compiler_4.5.1         mgcv_1.9-3             vctrs_0.6.5
## [10] pkgconfig_2.0.3        crayon_1.5.3           fastmap_1.2.0
## [13] backports_1.5.0        labeling_0.4.3         utf8_1.2.5
## [16] ggstance_0.3.7         promises_1.3.2         rmarkdown_2.29
## [19] tzdb_0.5.0             nloptr_2.2.1           bit_4.6.0
## [22] xfun_0.52              later_1.4.2            broom_1.0.8
## [25] parallel_4.5.1         R6_2.6.1               gap.datasets_0.0.6
## [28] stringi_1.8.7          qgam_2.0.0             RColorBrewer_1.1-3
## [31] car_3.1-3              boot_1.3-31            estimability_1.5.1
## [34] Rcpp_1.0.14            iterators_1.0.14       knitr_1.50
## [37] zoo_1.8-14             parameters_0.25.0      httpuv_1.6.16
## [40] Matrix_1.7-3           splines_4.5.1          timechange_0.3.0
## [43] tidyselect_1.2.1       rstudioapi_0.17.1      abind_1.4-8
## [46] yaml_2.3.10            MuMIn_1.48.11          doParallel_1.0.17
## [49] codetools_0.2-20       lattice_0.22-7         plyr_1.8.9
## [52] bayestestR_0.15.3      shiny_1.10.0           withr_3.0.2
## [55] evaluate_1.0.3         marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2          gap_1.6                carData_3.0-5
## [61] foreach_1.5.2          stats4_4.5.1           reformulas_0.4.1
## [64] insight_1.2.0          generics_0.1.4         vroom_1.6.5
## [67] rprojroot_2.0.4        hms_1.1.3              scales_1.4.0
## [70] minqa_1.2.8            xtable_1.8-4           glue_1.8.0
## [73] tools_4.5.1            data.table_1.17.2      lme4_1.1-37
## [76] mvtnorm_1.3-3          grid_4.5.1             rbibutils_2.3
## [79] datawizard_1.1.0       nlme_3.1-168           Rmisc_1.5.1
## [82] performance_0.13.0     beeswarm_0.4.0         vipor_0.4.7
## [85] Formula_1.2-5          cli_3.6.5              gtable_0.3.6
## [88] digest_0.6.37          pbkrtest_0.5.4         TH.data_1.1-3
## [91] farver_2.1.2           htmltools_0.5.8.1      lifecycle_1.0.4
## [94] mime_0.13              bit64_4.6.0-1          dotwhisker_0.8.4
## [97] MASS_7.3-65
```